A4.7 Access permissions

AXI provides access permissions signals that can be used to protect against illegal transactions:

- **ARPROT[2:0]** defines the access permissions for read accesses
- **AWPROT[2:0]** defines the access permissions for write accesses.

The term **AxPROT** refers collectively to the **ARPROT** and **AWPROT** signals.

Table A4-6 shows the **AxPROT[2:0]** encoding.

Table A4-6 Protection encoding

AxPROT	Value	Function
[0]	0	Unprivileged access
	1	Privileged access
[1]	0	Secure access
	1	Non-secure access
[2]	0	Data access
	1	Instruction access

The protection attributes are:

Unprivileged or privileged

An AXI master might support more than one level of operating privilege, and extend this concept of privilege to memory access. **AxPROT[0]** identifies an access as unprivileged or privileged.

Note

Some processors support multiple levels of privilege, see the documentation for the selected processor to determine the mapping to AXI privilege levels. The only distinction AXI can provide is between privileged and unprivileged access.

Secure or Non-secure

An AXI master might support Secure and Non-secure operating states, and extend this concept of security to memory access. **AxPROT[1]** identifies an access as Secure or Non-secure.

Note

This bit is defined so that when it is asserted the transaction is identified as Non-secure. This is consistent with other signaling in implementations of the ARM Security Extensions.

Instruction or data

This bit indicates whether the transaction is an instruction access or a data access.

The AXI protocol defines this indication as a hint. It is not accurate in all cases, for example, where a transaction contains a mix of instruction and data items. This specification recommends that a master sets **AxPROT[2]** LOW, to indicate a data access unless the access is specifically known to be an instruction access.

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